NIGHT DRIVE? NOT YET!

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Background

LiDAR

- Uses lasers mounted onto a vehicle to create a point-cloud representation of the area around the car

NuScenes [2]

- A multimodal dataset for autonomous driving (LiDAR and camera)
- Contains day and night scenes
- Collected in Boston (USA) and Singapore.



Fig 1: Example of NuScenes LiDAR data [2]

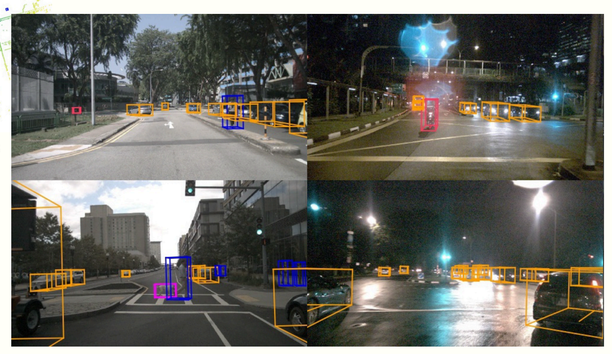


Fig 2: Example of detection on NuScenes

Goal

Detection models using **LiDAR** data **perform significantly worse on night scenes** compared to day scenes [1]. This happens for both LiDAR-only and multimodel models. Note that since LiDAR uses lasers, the reason that the models perform worse at night is not immediately obvious.

Investigation focus:

- Confirmation of the issue
- Dataset exploration
- Exploring the issue using experiments

Confirmation of the Issue

mAP%	Modality	Theirs		Ours	
		Day	Night	Day	Night
CenterPoint	LiDAR	62.8	35.4	59.6	33.9
TransFusion-L	LiDAR	-	-	64.7	37.5
BevFusion	LiDAR & Camera	68.5	42.8	67.8	42.2

Fig 3: Reproduction of experimental results from [1] on NuScenes validation set. We are able to reproduce the failing of models on night data, for both LiDAR-based models and multimodal models.

Hypothesis 1 (continued)

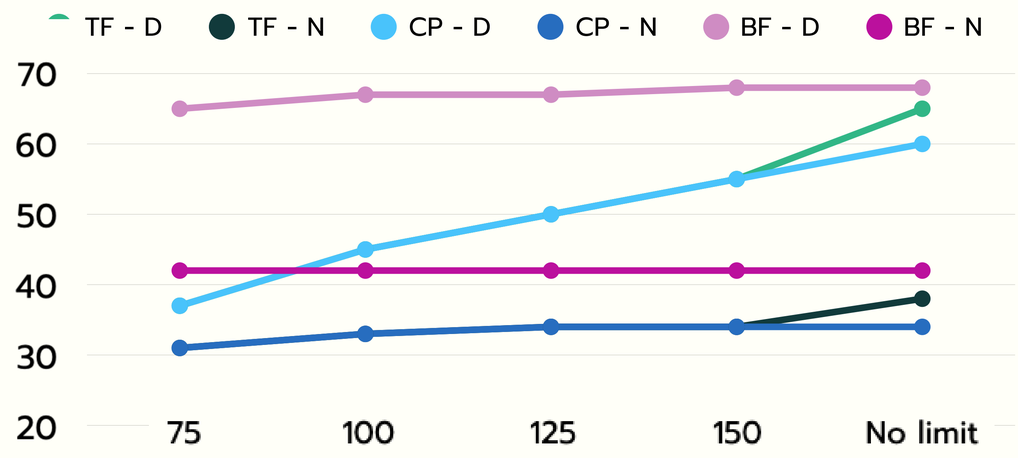


Fig 5: mAP% for day and night scenes with a limited number of predicted bounding boxes.. TF - TransFusion-L, CP - CenterPoint, BF - BevFusion. N denotes Night, D denotes Day.

Dataset Statistics

88% of the training data scenes are day scenes

An average day scene contains 2708 bounding boxes as compared to 737 in night scenes

Fig 4b: Comparison of day and night data in NuScenes.

1. There are significantly fewer night scenes than day
2. The night scenes have significantly fewer bounding boxes
3. The class balance is different for day and night scenes for all common classes

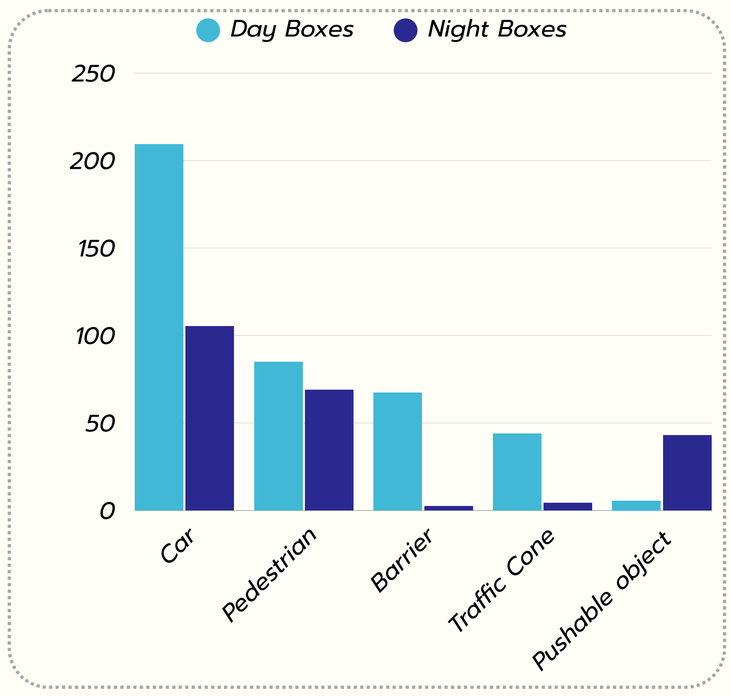


Fig 4a: Comparison of the number of bounding boxes for day and night scenes in NuScenes training data. Average number of bounding boxes per scene for the 4 most occuring day classes and the 'pushable object' class are shown.

Hypothesis 2: Data Imbalance

- NuScenes contains night data, but it is imbalanced (see figure 4b).
- To test, we retrained CenterPoint on a subsample of the training data containing 50% day data and 50% night data.
- Due to time constraints, we were only able to train 9 epochs. The original CenterPoint model was trained for 20 epochs.
- Though performance overall is lower, we see the day and night performance is closer together

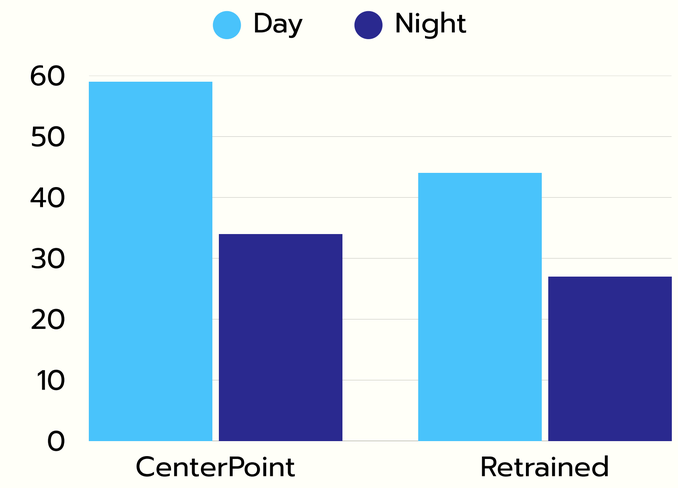


Fig 6: Comparison of day original CenterPoint performance and CenterPoint performance after balancing day and night scenes of the dataset.

Hypothesis 1: Wrong Boxes

- Following dataset statistics, our hypothesis is that the model has learned to predict many boxes for all scenes
- This would explain the low mAP for night scenes
- To test: we limit the number of bounding boxes predicted. If too many boxes are predicted for night, limiting this number would increase mAP.
- From Figure 5, we see that this does not occur. So, we move on to Hypothesis 2.

Conclusion

- **Confirmation of issue:** We replicated that three LiDAR detection models perform worse at night.
- **Dataset exploration:** NuScenes data is imbalanced, with 88% of scenes during the day. Day scenes have significantly more bounding boxes and different class balances compared to night scenes.
- **Issue exploration:** We tested two hypotheses to understand the performance drop at night: models predicting too many bounding boxes, which was not supported, and data imbalance. Retraining the CenterPoint model with balanced day and night data narrowed the performance gap, indicating **data imbalance as a key factor**.

Cai, R., Yang, B., Qi, C. R., & Xiao, J. (2023). ObjectFusion: Multi-modal 3D Object Detection with Object-Centric Fusion. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) 2023
Caesar, H., Bankiti, V., Lang, A. H., Vora, S., Liong, V. E., Xu, Q., Krishnan, A., Pan, Y., Baldan, G., & Beijbom, O. (2020). nuScenes: A Multimodal Dataset for Autonomous Driving. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 11621-11631).
Jaritz, M., Vu, T.-H., de Charette, R., Wirbel, E., & Perez, P. (2020). xMUDA: Cross-Modal Unsupervised Domain Adaptation for 3D Semantic Segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (pp. 12605-12614).